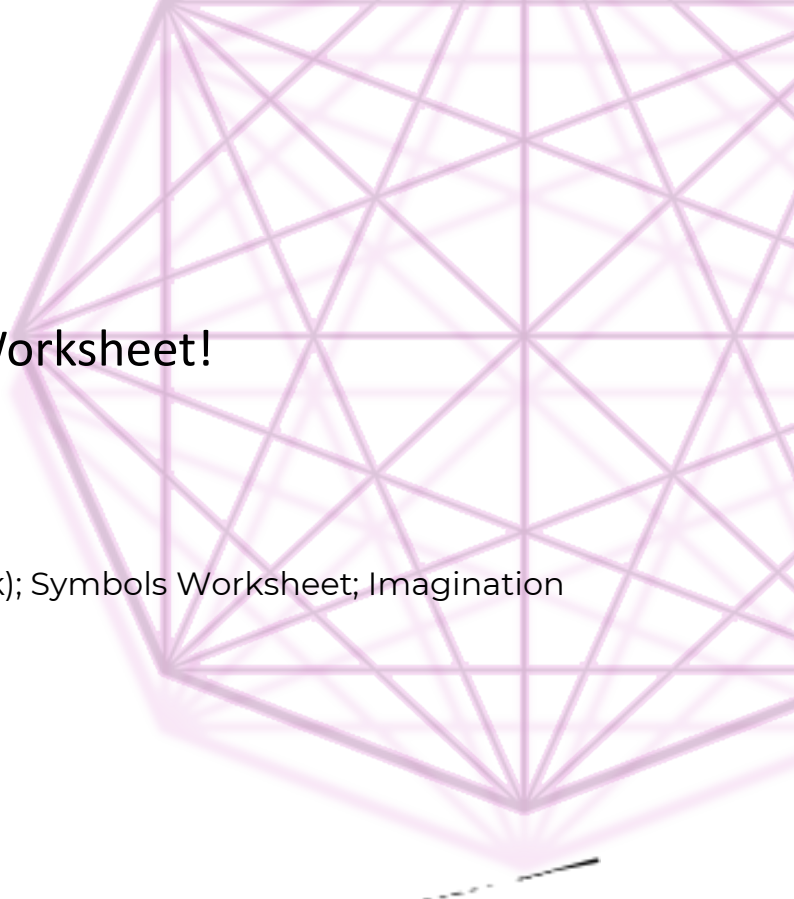




Mystic Math:

Sacred Shapes & Symbols Worksheet!

Course: SuperheroSTEM:

Materials: A pencil & paper (notebook); Symbols Worksheet; Imagination

Created by: Ryan Saelens



MISSION BRIEFING:

Welcome, Initiate.

For centuries, **the knowledge of shapes has been more than simple geometry** — it has been the **language of power, design, and hidden order**. Across ancient civilizations, symbols were created **not for decoration**, but for **precision**. For **purpose**. For **protection**.

These symbols — **circles, triangles, and other polygons** woven together — form the core of **sacred geometry**—where **magic and math intersect**.

You've been chosen to enter this forgotten tradition.

Your objective is clear:

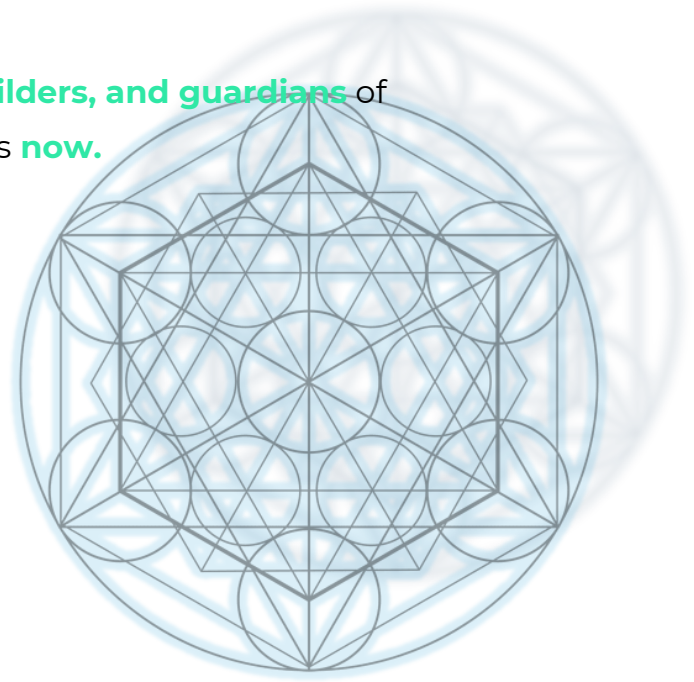
- **Study** the elemental forms: **the building blocks of all magical design**.
- **Decode** ancient seals and diagrams, **identifying the shapes hidden within**.
- **Use** geometry to **create your own symbol** — one that carries meaning and intent.
- **Complete** final tests of recognition, logic, and design to **prove mastery**.

This is no mere **math course**.

This is a training ground for **thinkers, builders, and guardians** of forgotten knowledge. Your **journey** begins **now**.

SKILLS TO MASTER:

- 🔍 Geometric Inquiry
- 🧠 Symbol Analysis
- 🎨 Creative Design
- 🎓 Mathematical Communication
- 📈 Pattern Recognition



LEARNING GOALS:

- **Identify and classify** 2D shapes and polygons by **sides, angles, and symmetry**
- **Recognize and analyze** shapes within **complex, symbolic designs, and Nature**
- **Construct** original symbols using **geometric shapes with intention**
- **Apply** shape recognition through visual tasks like **color-coding and design**
- **Use** precise mathematical vocabulary to **describe and compare shapes**
- **Understand** how geometry connects to **symbols, structures in Nature, and ancient knowledge**

So, **WHAT** exactly is **Geometry**?

Imagine the world as a **giant puzzle made of shapes**. **Geometry** is the **special language we use to understand those shapes** — from the **tiniest triangle** to the **largest circle** — and how they **fit together**.

Broadly speaking, **geometry** is the **study of points, lines, angles and surfaces**. Geometry helps us **see patterns in Nature**, design incredible

buildings, create powerful symbols, and even **understand how superheroes move through space!**

Think of geometry as **the magic behind the scenes** — the **rules** and **secrets that make everything** from spiderwebs to star maps **possible.**

So, whether you're **drawing a symbol or decoding an ancient seal,** geometry is your key to unlocking **the hidden order of the universe.**

Ready to become a **master of shapes and symbols?** **Let's dive in!**

Module 1: The Building Blocks of Magical Symbols!

Familiarize yourself with the **fundamental 2D shapes and polygons** that form the **foundation of all magical symbols.**

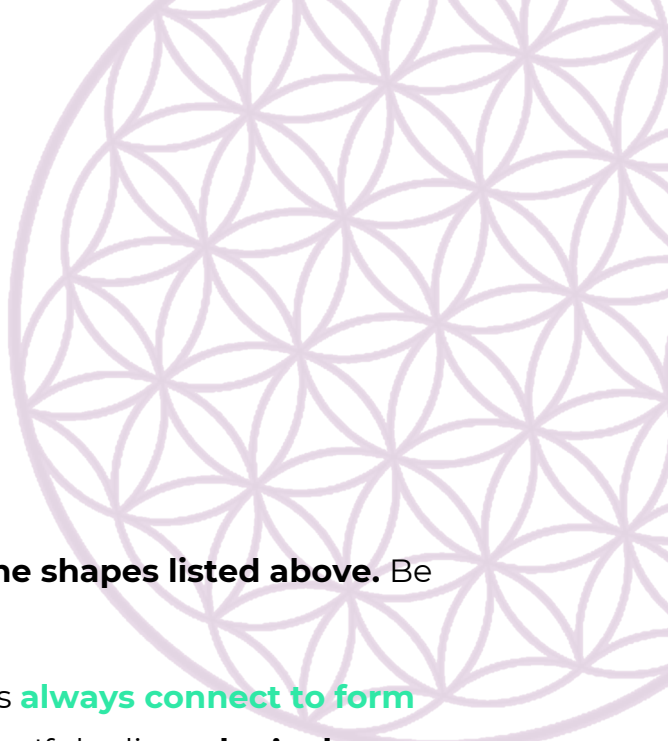
The **most fundamental** shapes, otherwise known as **primitive shapes,** are simply the following:

1. Triangle
2. Square
3. Rectangle
4. Circle

What is a *POLYGON* then?

A **polygon** is a flat, **closed shape** made by **connecting straight line segments end to end.** (QUICK! Is a circle a polygon?)

- Each straight line is called a **side.**
- The points **where the sides meet** are called **vertices (think of them like corners).**
- **Polygons** can have **any number of sides** — from triangles with 3 sides, all the way up to shapes with many sides! **Here is a list of polygons with sides from 3 to 10:**
 - **Triangle** — 3 sides
 - **Quadrilateral** — 4 sides (includes squares, rectangles, trapezoids)
 - **Pentagon** — 5 sides

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- **Hexagon** — 6 sides
 - **Heptagon** — 7 sides
 - **Octagon** — 8 sides
 - **Nonagon** — 9 sides
 - **Decagon** — 10 sides

On a separate piece of paper, **draw each of the shapes listed above**. Be sure to **label each shape**.

What makes a polygon **special** is that its sides **always connect to form one continuous loop, with no gaps or curves**. If the lines **don't close together** to make a shape, **it isn't a polygon**.

Regular vs Irregular Polygons – WHAT'S the DIFFERENCE?

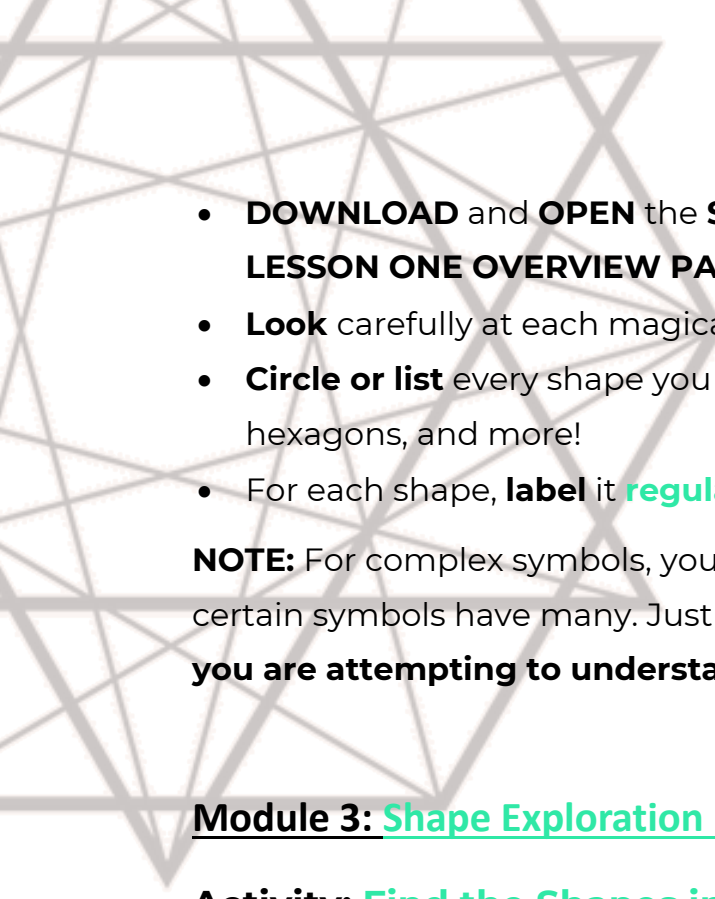
- **Regular polygons** have all their sides **the same length** and all their angles **the same size**.
 - Can you think of an example of a **regular polygon**?
 - **Draw your example in your notes.**
- **Irregular polygons** have sides and angles that **are not all the same size**.
 - Can you think of an example of an **irregular polygon**?
 - **Draw your example in your notes.**

Module 2: Shape Exploration Exercise #1

Now that you've met the basic shapes, **it's time to become a *Shape Detective!*** Magical symbols are built from these shapes — **your mission is to find them hidden inside**.

Activity: Find the Shapes in the Symbols!

Instructions:

- 
- **DOWNLOAD** and **OPEN** the **SYMBOLS WORKSHEET** found on the **LESSON ONE OVERVIEW PAGE**
 - **Look** carefully at each magical symbol! There are **five**.
 - **Circle or list** every shape you can find — triangles, circles, squares, hexagons, and more!
 - For each shape, **label** it **regular** or **irregular**.

NOTE: For complex symbols, you **DO NOT** have to label every shape—as certain symbols have many. Just **label a handful**—enough to show that **you are attempting to understand the assignment**.

Module 3: Shape Exploration Exercise #2

Activity: Find the Shapes in Nature!

Instructions:

- **Look carefully at objects in Nature** such as, trees, fields, flowers, rocks, etc.!
- **Make a list of objects** along with the shapes and/or symbols (**use the symbols from the symbols worksheet**) **that help create it!**
- For each shape, **label** it **regular** or **irregular**.

Guiding Questions For Above Exercises!

- Do you see any **triangles**?
- Can you find any **circles or semicircles**?
- Are the shapes all the **same size or different**?
- Which shapes are **regular**? Which are **irregular**?
- Can you spot any polygons with **5 or more sides**?
- Which symbol was the **most complex**?
- Did you find any of the magical symbols **in Nature**? Please list them.

Module 4: Symbol Creation Challenge!

You've explored the building blocks of magical symbols, as well as how these symbols hide in Nature. **Now it's your turn to become a symbol maker!** Using circles, triangles, squares, and other polygons, **you'll create a symbol (or two—shoot, make three if you want!) that could hold power and tell a story!**

Instructions:

1. **Choose your shapes:** Pick **4-6 shapes** from the set to **use in your symbol.**
2. **Arrange your shapes:** Think about how the shapes fit together. **Overlap, rotate, or place them side-by-side** to create an **interesting design.**
3. **Think about meaning:** What might your symbol represent? **Power, protection, wisdom?** You can **give your symbol a name!**
4. **Draw and color:** Sketch your symbol and **add colors if you want!**
5. **Describe your symbol:** Write a short explanation of the shapes you used and **what your symbol means.**

Mission Checklist:

- ▲ Shapes and polygons **reviewed**
- 🧠 Hidden shapes **identified** within magical symbols – **and** Nature
- 🌀 Custom symbol(s) **designed** using geometric forms
- ✎ Explanation **written** for your symbol's design and meaning
- ✅ Final symbol(s) **polished** and ready to **present** or **share!**

Image Credit: I do not own the background images. Though these symbols are sacred symbols that can be found in reality, the ones' being used for this worksheet are products of Vexels.com.